Math 9 - Similar Polygons and Transformations



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

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Teacher's name: _____

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Last Updated: April 17, 2025

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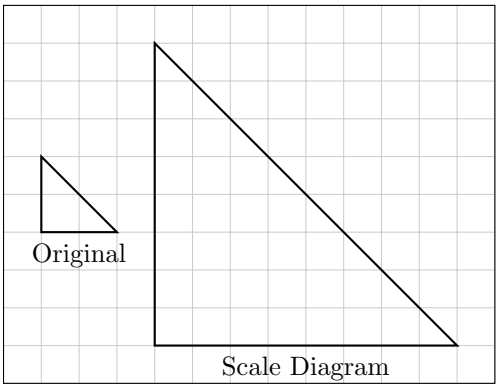
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1 Scale Factors

Scale factors can either **enlarge** an object or **reduce** an object.

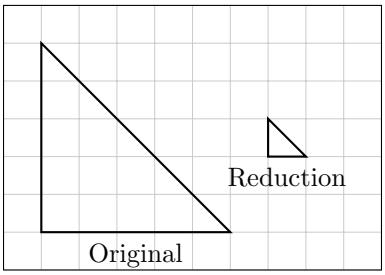
A scale factor that is greater than 1 is considered an **enlargement**. All dimensions of the enlarged image will be greater than the original object.

Example 1.1 Identify the scale factor of the enlargement below.



A scale factor that is less than 1 (greater than 0) is considered a **reduction**. All dimensions of the reduced image will be less than the original object. These scale factors are usually represented as a fraction or a decimal between 0 and 1.

Example 1.2 Identify the scale factor of the reduction below.



Regardless of an image being an enlargement or a reduction, the scale factor can be calculated using the following.

$$\text{Scale Factor} = \frac{\text{Image Length}}{\text{Object length}}$$

This only holds true when the two objects are **proportional** to each other. More on this later.

Example 1.3 Complete the following table.

Object Length	Image Length	Scale Factor	Enlargement or Reduction
12 cm	24 cm		
14.7 cm		4	
25 cm	5 cm		
	48 cm	1.5	
65 cm		0.2	
	24 cm	0.4	

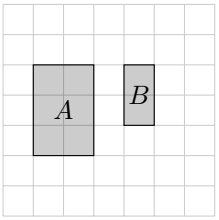
Example 1.4 A doll house is one thirtieth the actual size of a house. If a chair in the doll house is 2.75 cm, how tall is the actual chair it models?

Example 1.5 On a map, the distance between two cities is 7.5 cm. The map uses a scale of 1:500,000 (1 cm represents 500,000 cm in real life). What is the actual distance between the two cities in kilometers?

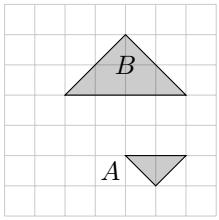
Practice Exercises

1. If A is the object and B is the image, state whether each of the following is an enlargement or a reduction.

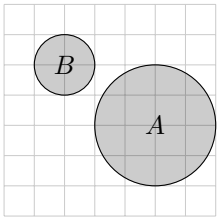
a)



b)

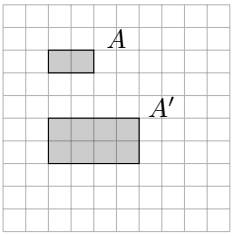


c)

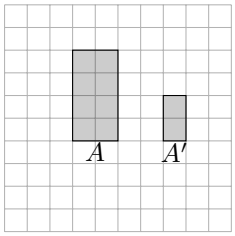


2. Find the scale factors where A is the original and A' is the image.

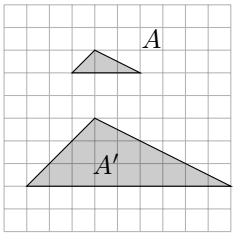
a)



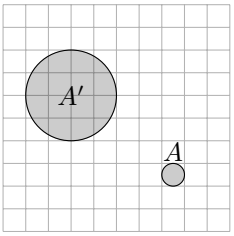
b)



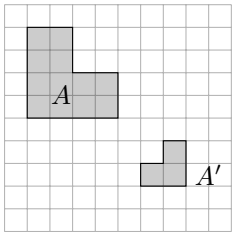
c)



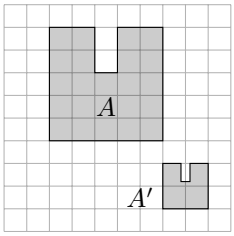
d)



e)



f)

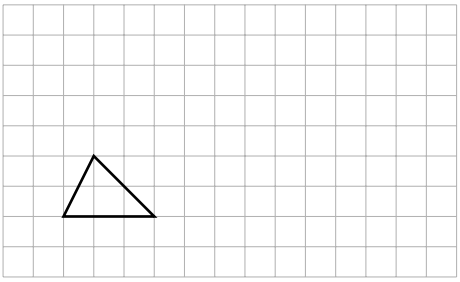


3. Complete the following table.

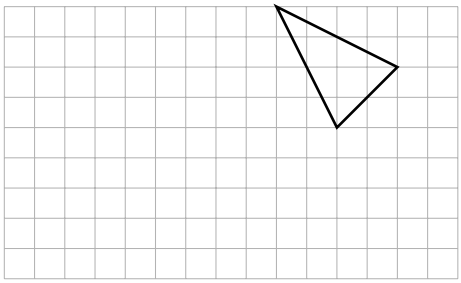
Object Length	Image Length	Scale Factor	Enlargement or Reduction
12 cm	36 cm		
10.1 cm		4	
54 cm		$\frac{1}{9}$	
	36 cm	0.5	

4. Draw each image magnified by a scale factor of 2 and by $\frac{1}{2}$.

a)

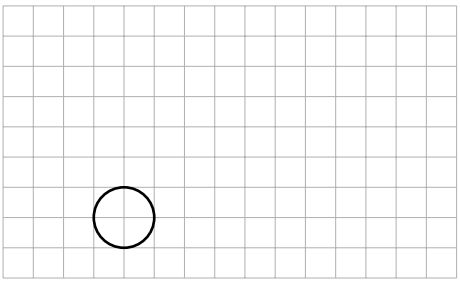


b)

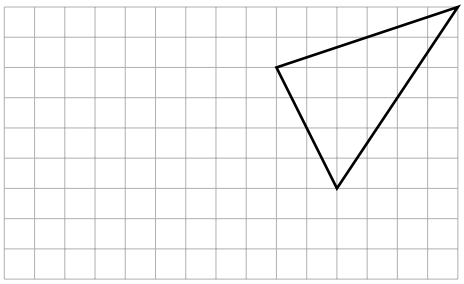


5. Graph each image as indicated.

a) Scale factor of 3



b) Scale factor of $\frac{1}{2}$



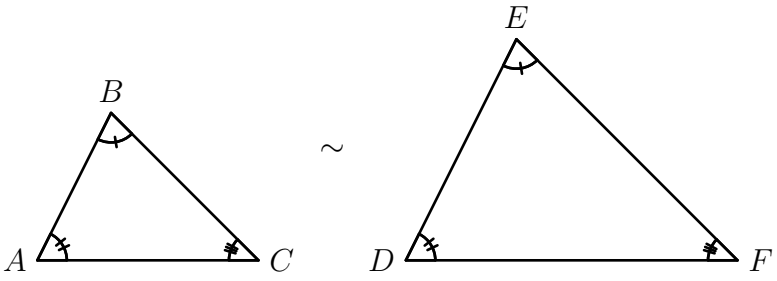
6. A portrait measures 50 cm by 35 cm. If the larger side of a reduction measures 20 cm,what is the length of the shorter side of the reduction?

7. A pole 3 m high has a shadow 5 m long. If a building is 66 m high, how long is the building's shadow?

2 Similar Triangles and Polygons

Triangles have a property that all their interior angles must sum to 180°.

Two triangles are similar if all their interior angles are the same. If triangles are similar, their sides are proportional.



The two triangles above are similar, that is $\triangle ABC \sim \triangle DEF$ and $\angle A = \angle D$, $\angle B = \angle E$, and $\angle C = \angle F$.

Because the triangles are similar, the ratios of the corresponding sides are equal. That is

$$AB : DE \qquad BC : EF \qquad AC : DF$$

will have the same ratio.

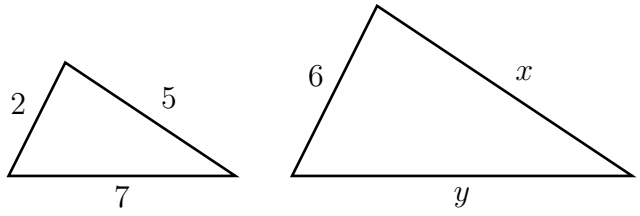
The key skill we use to solve for sides in similar triangles (and polygons) is setting up a proportional equation and solving.

Example 2.1 Solve the following equations.

a) $\dfrac{3}{x} = \dfrac{4}{9}$

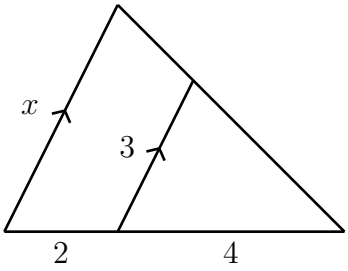
b) $\dfrac{x-1}{6} = \dfrac{x+1}{10}$

Example 2.2 Determine the length of x and y given that the two triangles below are similar.

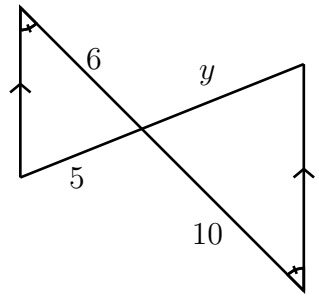


Some triangles are harder to identify. A suggestion is to draw them out separately. Similar to the previous example.

Example 2.3 Find x in



Example 2.4 Find y in



Example 2.5 Triangle A has angles 40° and 75°, and Triangle B has angles 65° and 40°. Are the two triangles similar? How do you know?

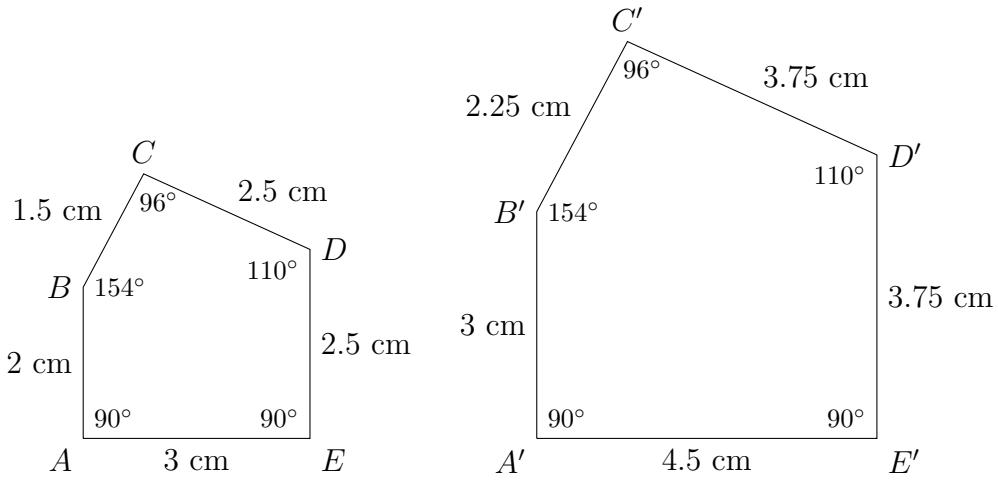
Similar Polygons

We approach similar polygons almost the same way as similar triangles. However, the requirement for similarity in polygons is not just equal angles.

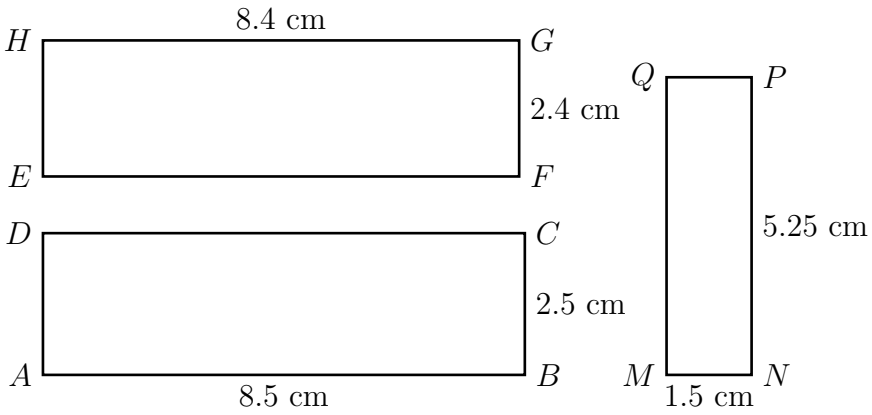
Example 2.6 Draw two quadrilaterals that have all angles equal but are not similar to each other.

For two polygons to be similar, all corresponding sides must have equal ratios or be proportional, and all angles must be the same.

Example 2.7 Here are two similar pentagons. Check that the corresponding sides have equal ratios.

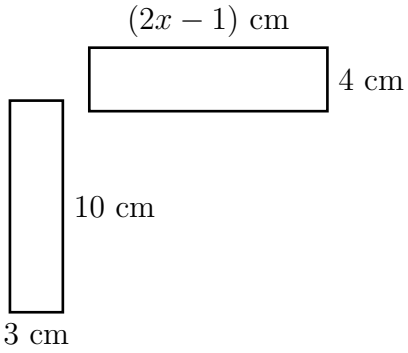


Example 2.8 Which pairs of rectangles below are similar?

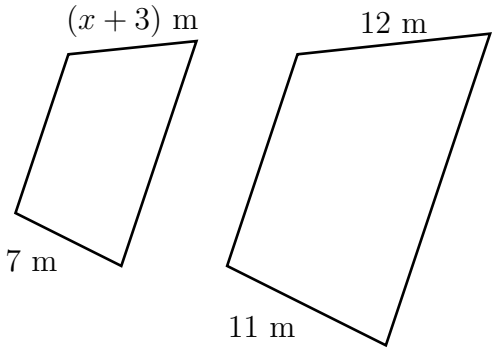


Example 2.9 Find the value of x given that the polygons are similar.

a)



b)



Practice Exercises

1. Find the value of x.

a) $\frac{x}{7} = \frac{3}{4}$

b) $\frac{x+3}{4} = \frac{9}{2}$

c) $\frac{9}{x} = \frac{5}{7}$

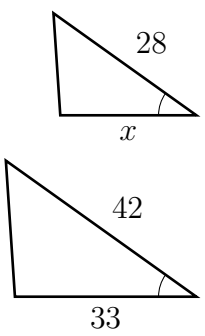
d) $\frac{x-4}{5} = \frac{x-3}{7}$

e) $\frac{x+4}{3} = \frac{x+8}{5}$

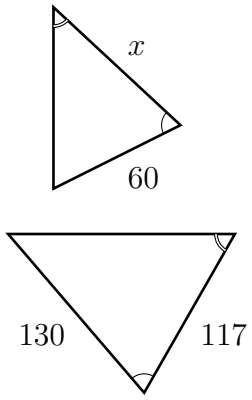
f) $\frac{x-2}{3} = \frac{x+4}{7}$

2. Find x, the unknown side in the following pairs of similar triangles.

a)

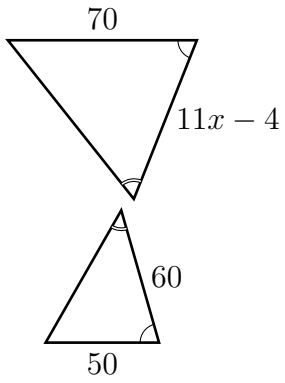


b)

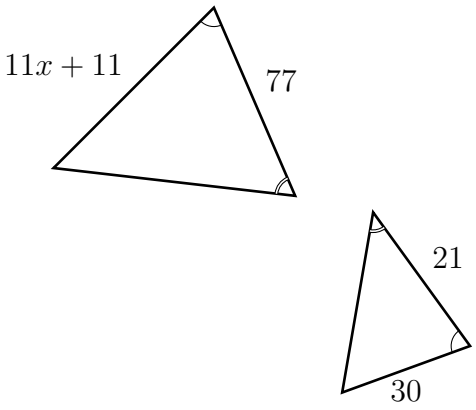


3. Solve for x

a)

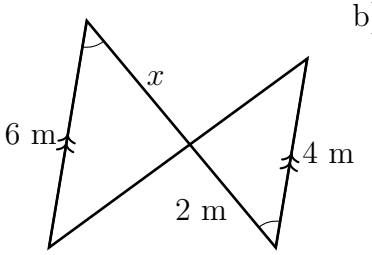


b)

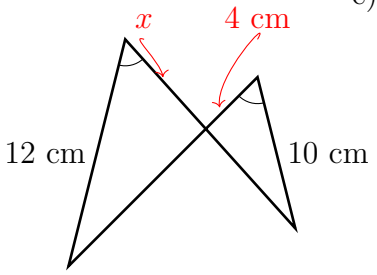


4. Find the unknowns in:

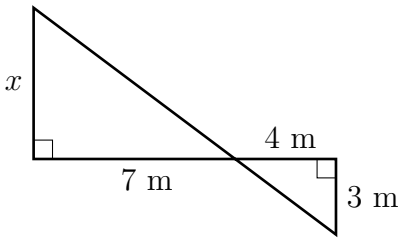
a)



b)

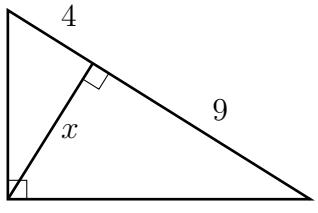


c)

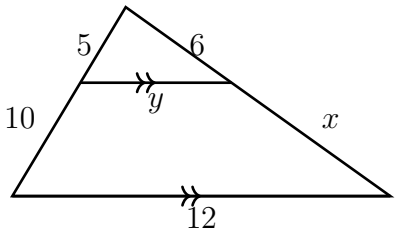


5. Solve for the unknown values.

a)



b)



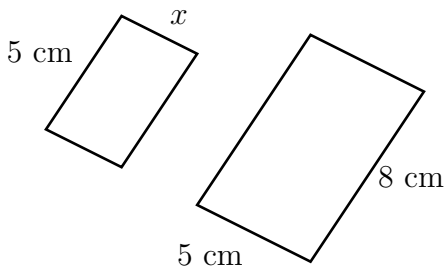
6. Sketch two quadrilaterals which:

   a) Have equal angles but are not similar.

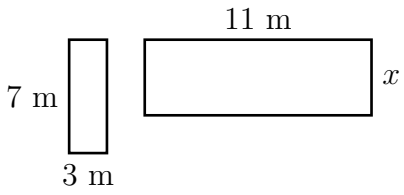
   b) Have sides in proportion, but are not similar.

7. Solve for x given that the figures are similar:

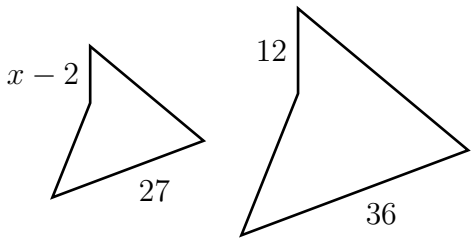
a)



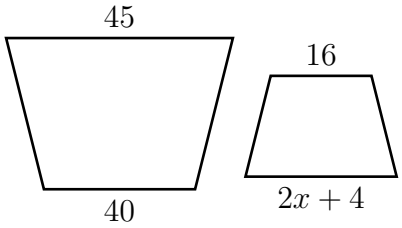
b)



c)

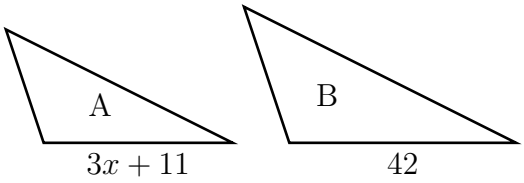


d)

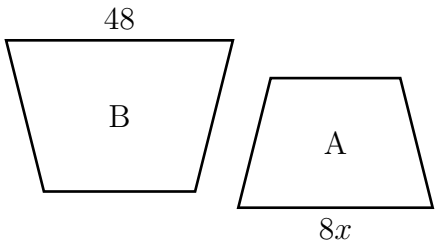


8. Given the scale factor ratio, solve for x.

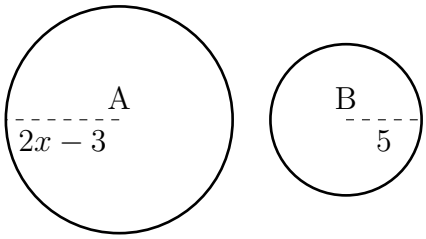
a) Scale factor from A to B = 5:6



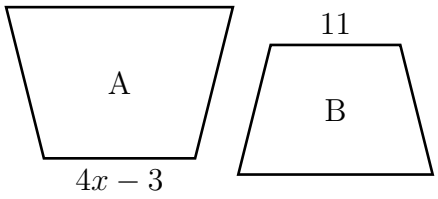
b) Scale factor from A to B = 6:7



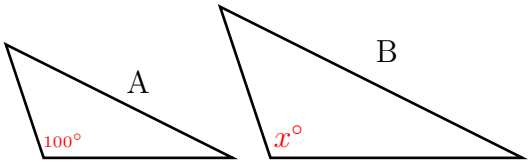
c) Scale factor from A to B = 3:2



d) Scale factor from A to B = 7:6



e) Scale factor from A to B = 3:4



3 Point Transformation

Translations

Transformations is a topic that will be revisited frequently in later years.

We will introduce some vocabulary, notation, and transformations on points.

A **translation** on a point is moving the point horizontally and/or vertically.

A common notation we use to show transformations is the **mapping notation**. Taking a point (x, y) and mapping it to a new point.

$$(x, y) \to (x + a, y + b)$$

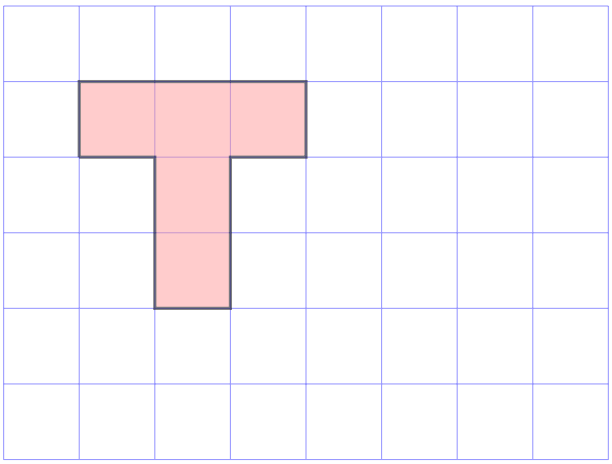
Example 3.1 Where will the point $(3, 4)$ be after a translation of 2 units left and 5 units down?

Example 3.2 State the translations given by the below mapping notations.

a) $(x, y) \to (x + 1, y - 6)$

b) $(x, y) \to (x - 3, y + 2)$

Example 3.3 Draw the below figure after a translation of 2 units to the right and 3 units down.



Reflections

A **reflection** over the x-axis or y-axis is when the point is "flipped" over. Its mapping notation is given by

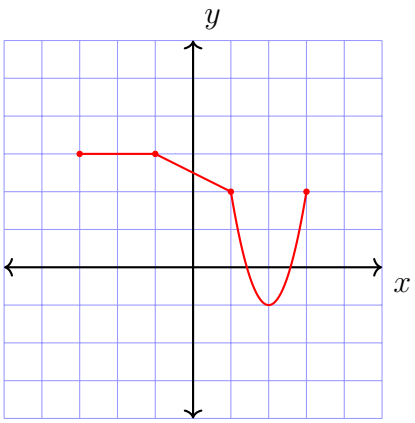
$$(x, y) \to (-x, y) \text{ for a reflection over the } y\text{-axis}$$
$$\text{and}$$
$$(x, y) \to (x, -y) \text{ for a reflection over the } x\text{-axis}$$

Example 3.4 Identify where the point $(3, -2)$ will be after a reflection in the y-axis.

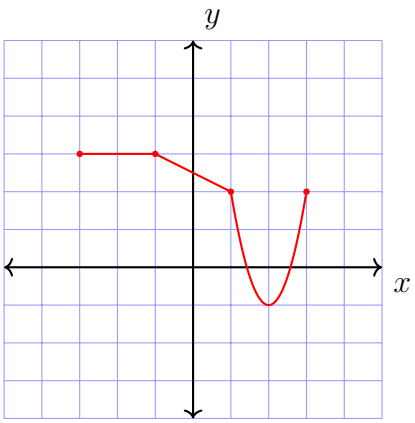
Example 3.5 Identify where the point $(3, -2)$ will be after a reflection in the x-axis.

Example 3.6 Draw the curve below after a reflection in the indicated axis.

a) y-axis



b) x-axis



Stretches

A **stretch** is essentially a scale factor for points. Depending on the stretch, we can classify them as an **expansion**, or a **compression**. Horizontal stretch is always with respect to the y-axis and vertical stretch is always with respect to the x-axis.

Its mapping notation is given by

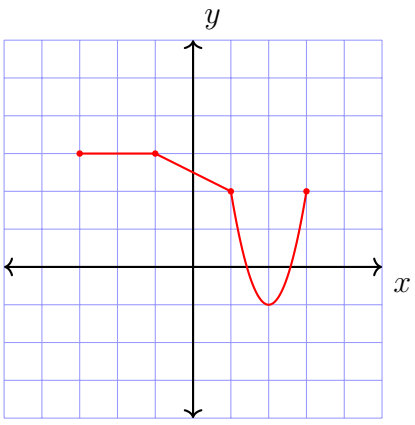
$$(x, y) \to (ax, y) \text{ for a horizontal stretch by a factor of } a$$
$$\text{and}$$
$$(x, y) \to (x, ay) \text{ for a vertical stretch by a factor of } a$$

Example 3.7 Identify where the point $(3, -2)$ will be after a horizontal stretch by a factor of $\frac{1}{3}$

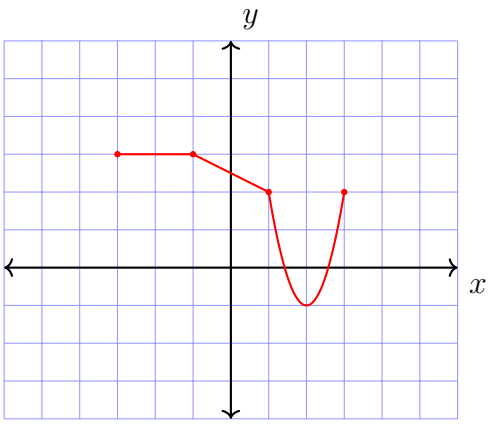
Example 3.8 Identify where the point $(3, -2)$ will be after a vertical stretch by a factor of 3.

Example 3.9 Draw the curve below after the indicated stretch.

a) Vertical cmpression by a factor of $\frac{1}{2}$



b) Horizontal stretch by a factor of 2



Example 3.10 Write the mapping notation on (x, y) given the series of transformations. Then use the point $(-2, 3)$ and identify where it is after the transformations.

a) Reflect over the y-axis. Translate 2 left and 4 up.

b) Vertical stretch by 2. Translate 3 right and 3 down.

Example 3.11 State the transformations given the following mapping. Then use the point $(-2, 3)$ and identify where it is after the transformations.

a) $(x, y) \to \left(2x - 1, \frac{1}{3}y\right)$

b) $(x, y) \to \left(\frac{1}{2}x + 1, -y + 3\right)$

Order of Transformations

Depending on the order of transformations, we may end up with different values.

Example 3.12 Compare the following two mapping notations by using the point $(2, 2)$ as the original point and see where the transformations takes the point. List the transformations represented by the mapping notation.

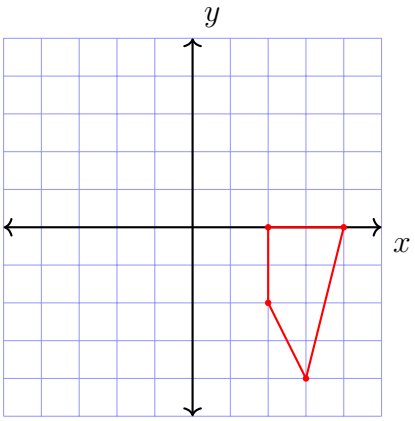
a) $(x, y) \to (2x + 1, -y - 1)$

b) $(x, y) \to (2(x + 1), -(y - 1))$

Practice Exercises

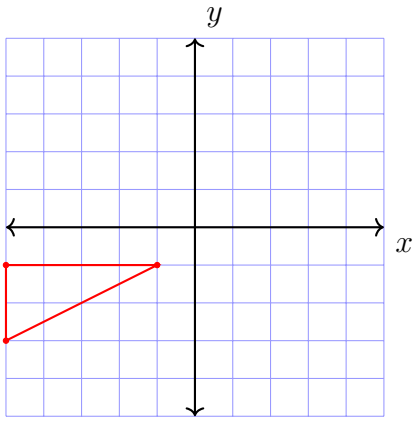
1. Graph the image of the figure after the transformations given. Write the mapping notation given by the transformation below.

a) Translation: 1 unit left



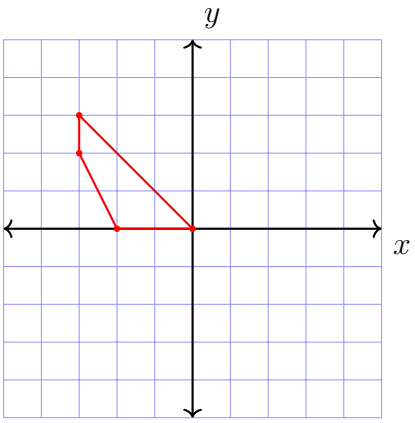
$(x, y) \rightarrow (\quad , \quad)$

b) Translation: 3 units right



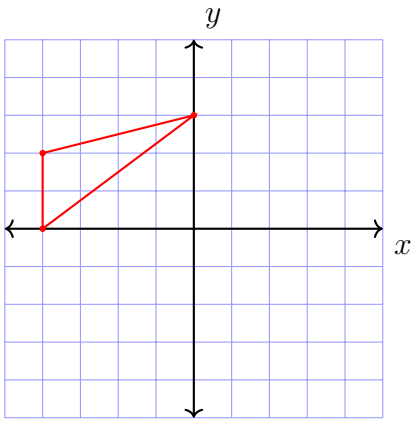
$(x, y) \rightarrow (\quad , \quad)$

c) Translation: 1 unit right and 2 units down



$(x, y) \rightarrow (\quad , \quad)$

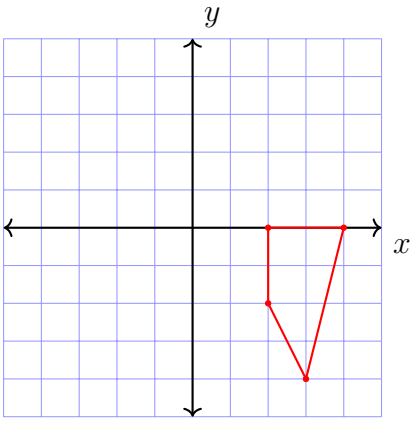
d) Translation: 1 unit right and 2 units down



$(x, y) \rightarrow (\quad , \quad)$

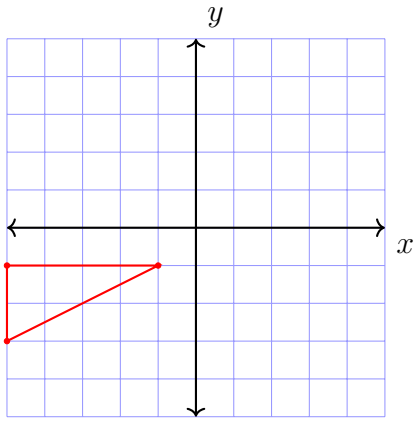
2. Graph the image of the figure after the transformations given. Write the mapping notation given by the transformation below.

a) Reflection over x-axis



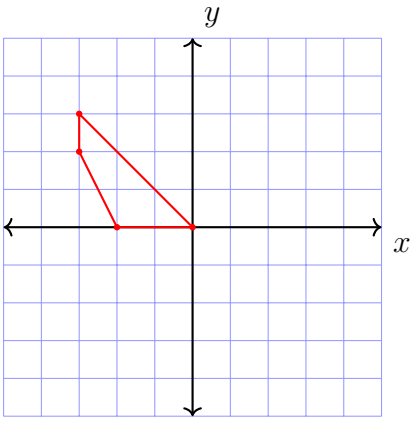
$(x, y) \rightarrow (\quad , \quad)$

b) Reflection over y-axis



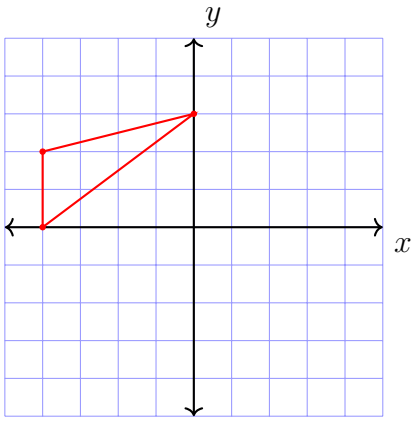
$(x, y) \rightarrow (\quad , \quad)$

c) Reflections over x-axis and then y-axis



$(x, y) \rightarrow (\quad , \quad)$

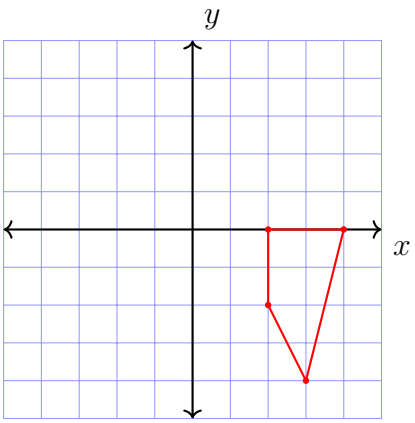
d) Reflections over y-axis and then x-axis



$(x, y) \rightarrow (\quad , \quad)$

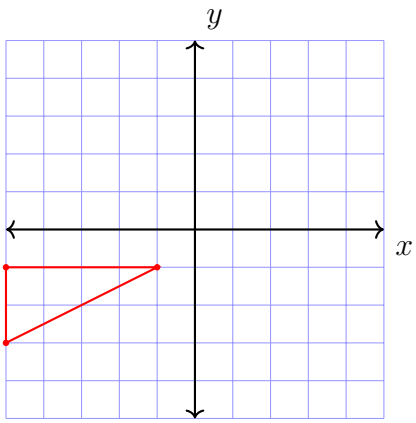
3. Graph the image of the figure after the transformations given. Write the mapping notation given by the transformation below.

a) Horizontal stretch by a factor of $\frac{1}{2}$



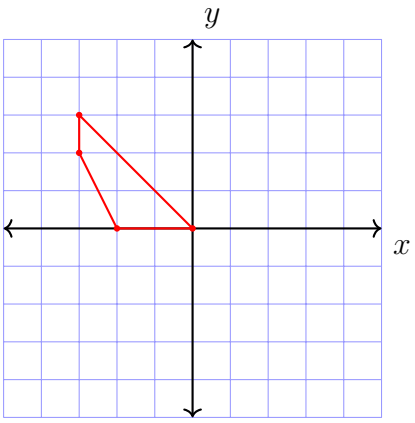
$(x, y) \rightarrow (\quad , \quad)$

b) Vertical stretch by a factor of $\frac{1}{2}$



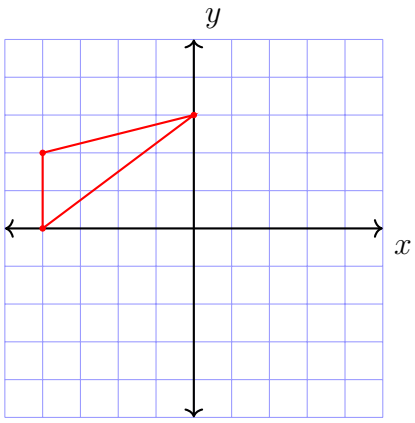
$(x, y) \rightarrow (\quad , \quad)$

c) Horizontal stretch by a factor of $\frac{3}{2}$



$(x, y) \rightarrow (\quad , \quad)$

d) Vertical stretch by a factor of $\frac{3}{2}$



$(x, y) \rightarrow (\quad , \quad)$

4. State the transformation(s) indicated by the mapping notation.

a) $(x, y) \to (x + 3, y - 2)$

b) $(x, y) \to (x - 4, 2y)$

c) $(x, y) \to (x + 3, y - 2)$

d) $(x, y) \to (x - 4, 2y)$

e) $(x, y) \to (x + 3, y - 2)$

f) $(x, y) \to (x - 4, 2y)$

g) $(x, y) \to (x + 3, y - 2)$

h) $(x, y) \to (x - 4, 2y)$

i) $(x, y) \to (x + 3, y - 2)$

j) $(x, y) \to (x - 4, 2y)$

5. Write the mapping notation indicated by the transformations. Transformations are listed in order.

a) 2 units up
3 units left

b) Reflection over y-axis
6 units up

c) Reflection over x-axis
4 units right

d) Horizontal stretch by a factor of 3
Reflection over x-axis

e) Vertical stretch by a factor of $\frac{2}{7}$
Horizontal stretch by a factor of $\frac{7}{2}$

f) Horizontal stretch by a factor of 3
Vertical stretch by a factor of 4

g) Horizontal stretch by a factor of 2
5 units right
Reflection over x-axis

h) Vertical stretch by a factor of $\frac{1}{3}$
2 units down
Reflection over y-axis
3 units left

i) 4 units right
Horizontal stretch by a factor of $\frac{2}{5}$
3 units up
Vertical stretch by a factor of 5

j) 3 units down
Reflection over x-axis
2 units left
Reflection over y-axis

4 Reasoning and Analyzing Problems

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

Example 4.1 For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) A scale factor of 2 represents an enlargement.

b) Two triangles are similar if they have one equal angle.

c) Two triangles are similar if they have two equal angles.

d) If two polygons with equal angles, then they are similar.

e) $(x, y) \to (x + a, y)$ represents a translation to the right

f) The point (a, b) is vertically stretched by a factor of 2, the resulting point is (a, b).

Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) The point (a, b) is horizontally stretched by a factor of 2, the resulting point is (a, b).

   (b) $(x, y) \to (x, y - b)$ represents a vertical translation down.

   (c) Any two equilateral triangles are similar.

   (d) If two triangles are both isosceles triangles, then they are similar.

   (e) If a quadrilateral $ABCD$ is an enlargement of another quadrilateral $EFGH$, then the corresponding angles in $ABCD$ are larger than then angles in $EFGH$.

   (f) For non-zero values a, b, c, d, if $\dfrac{a+b}{c} = \dfrac{d}{a+b}$, then $a^2 + b^2 = cd$.

   (g) For any point (a, b), a reflection over the x-axis and then left 3 units results in $(a - 3, -b)$.

Answer Key

Section 1 - Exercises

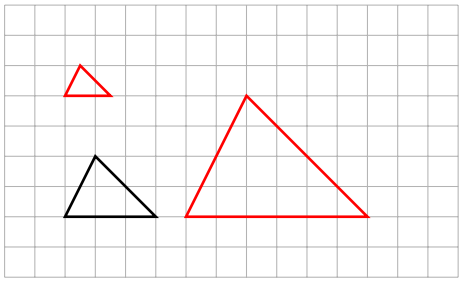
1. a) Reduction b) Enlargement c) Reduction

2. a) $S.F = 2$ b) $S.F = \frac{1}{2}$ c) $S.F = 3$
   d) $S.F = 4$ e) $S.F = \frac{1}{2}$ f) $S.F = \frac{2}{5}$

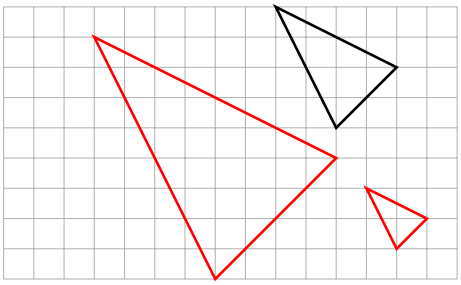
3.

Object Length	Image Length	Scale Factor	Enlargement or Reduction
12 cm	36 cm	3	Enlargement
10.1 cm	40.4 cm	4	Enlargement
54 cm	6 cm	$\frac{1}{9}$	Reduction
72 cm	36 cm	0.5	Reduction

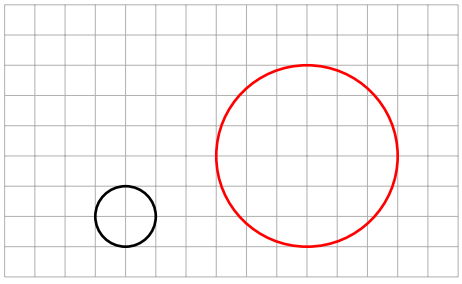
4. a)



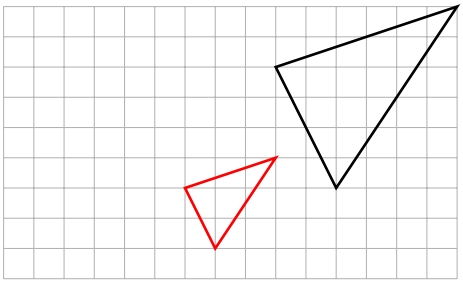
b)



5. a)



b)



6. The length of the shorter side is 14 cm.

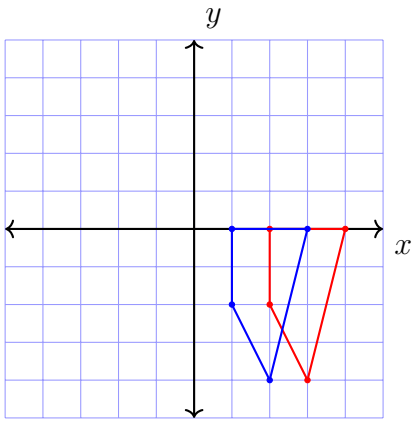
7. The building's shadow is 110 m.

Section 2 - Exercises

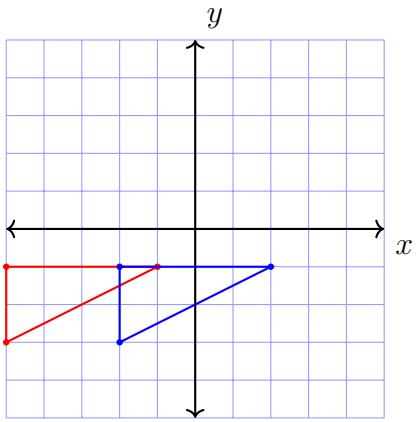
1. a) $x = \frac{21}{4}$ or 5.25 b) $x = 15$ c) $x = \frac{63}{5}$ or 12.6
   d) $x = \frac{13}{2}$ or 6.5 e) $x = 2$ f) $x = \frac{13}{2}$ or 6.5
2. a) $x = 22$ b) $x = 54$
3. a) $x = 8$ b) $x = 9$
4. a) $x = 3$ m b) $x = \frac{24}{5}$ cm or 4.8 cm c) $\frac{21}{4}$ m or 5.25 m
5. a) $x = 6$ b) $x = 12$, $y = 4$
6. a) A square and a rectangle. b) A rhombus and a square.
7. a) $x = \frac{25}{8}$ cm or 3.125 cm b) $x = \frac{33}{7}$ m
   c) $x = 11$ d) $x = 7$
8. a) $x = 8$ b) $x = \frac{36}{7}$
   c) $x = \frac{21}{4}$ or 5.25 d) $x = \frac{95}{24}$
   e) $x = 100°$

Section 3 - Exercises

1. a) b)

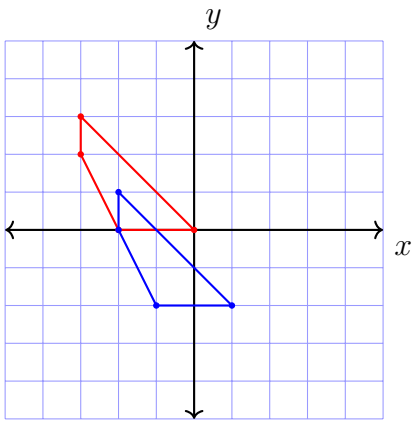


$(x, y) \to (x - 1, y)$



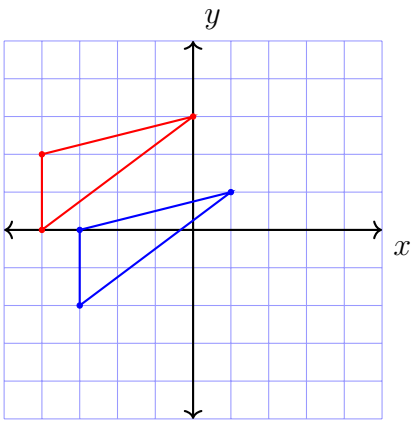
$(x, y) \to (x + 3, y)$

c)



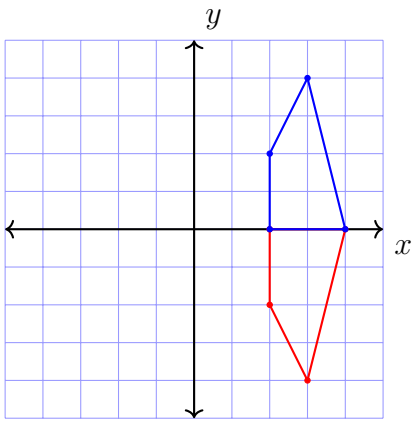
$(x, y) \to (x + 1, y - 2)$

d)



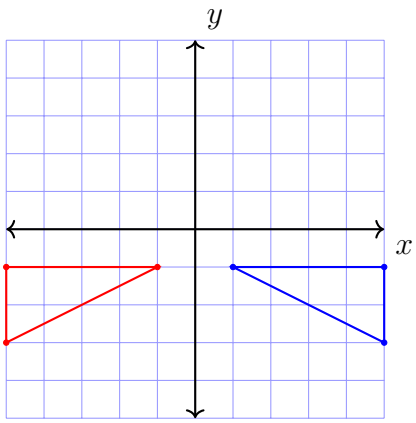
$(x, y) \to (x + 1, y - 2)$

2. a)



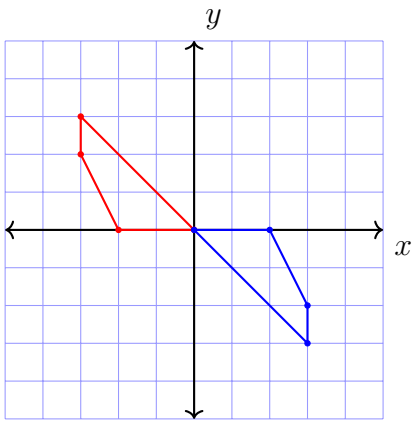
$(x, y) \to (x, -y)$

b)



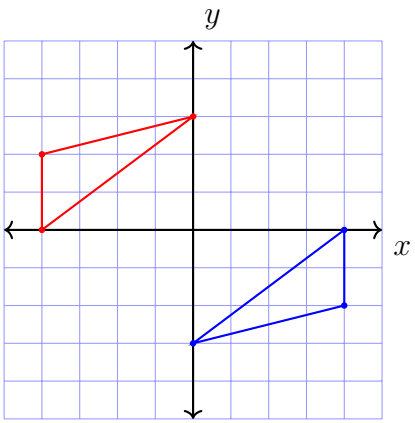
$(x, y) \to (-x, y)$

c)



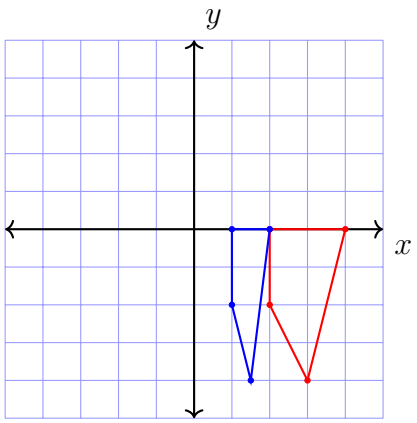
$(x, y) \to (-x, -y)$

d)



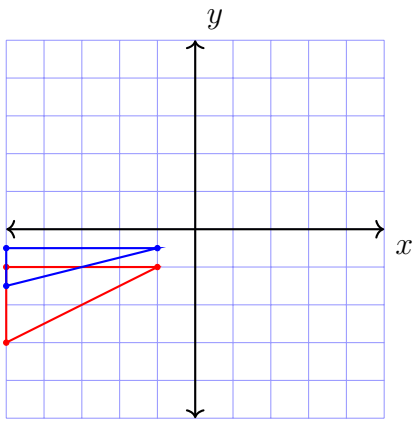
$(x, y) \to (-x, -y)$

3. a)



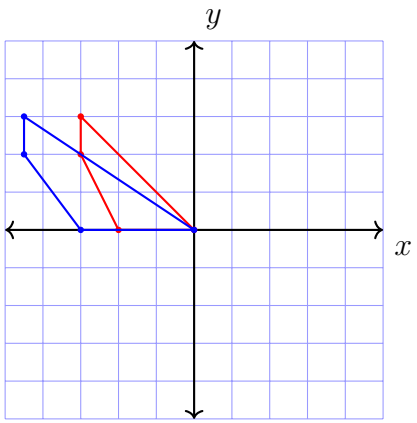
$(x, y) \to (\frac{1}{2}x, y)$

b)



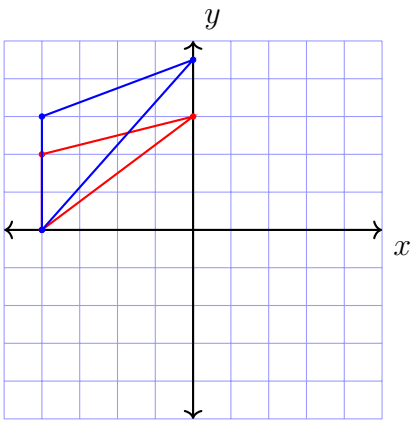
$(x, y) \to (x, \frac{1}{2}y)$

c)



$(x, y) \to (\frac{3}{2}x, y)$

d)



$(x, y) \to (x, \frac{3}{2}y)$

4. a) right 3 units
down 2 units

b) left 4 units
vertical stretch by 2

c) reflect over y-axis
up 4 units

d) horizontal stretch by 2
reflect over x-axis

e) horizontal stretch by $\frac{1}{4}$
up 1 unit

f) horizontal stretch by 3
vertical stretch by $\frac{2}{5}$

g) horizontal stretch by 2
right 1 unit
reflect over x-axis
down 3 units

h) reflect over y-axis
right 3 units
vertical stretch by $\frac{1}{2}$
up 3 units

i) right 1 unit
horizontal stretch by 2
down 3 units
reflect over x-axis

j) right 3 units
reflect over y-axis
up 3 units
vertical stretch by $\frac{1}{2}$

5. a) $(x, y) \to (x - 3, y + 2)$

b) $(x, y) \to (-x, y + 6)$

c) $(x, y) \to (x + 4, -y)$

d) $(x, y) \to (3x, -y)$

e) $(x, y) \to (\frac{7}{2}x, \frac{2}{7}y)$

f) $(x, y) \to (3x, 4y)$

g) $(x, y) \to (2x + 5, -y)$

h) $(x, y) \to (-x - 3, \frac{1}{3}y - 2)$

i) $(x, y) \to (\frac{2}{5}(x + 4), 5(y + 3))$

j) $(x, y) \to (-(x - 2), -(y - 3))$

Section 4 - Exercises

1. a) Sometimes true. For example, $(0, 1)$ if horizontally stretched is still $(0, 1)$. $(1, 1)$ would be $(2, 1)$ which is different.

   b) Sometimes true. If $b > 0$, then it will translate down. If $b \leq 0$ then it either doesn't translate vertically, or it can go up.

   c) Always true. All angles are $60°$, so the triangles are similar.

   d) Sometimes true. Similarity depends on the angles. So as long as the two isosceles triangles have equal angles, then they are similar. Otherwise, they are not similar.

   e) Never true. An enlargement preserves ratios and angles, so corresponding angles need to be the same.

   f) Never true. When cross-multiplying, we get $(a + b)^2 = cd$, and $(a + b)^2 = a^2 + 2ab + b^2$ and not $a^2 + b^2$.

   g) Always true. The mapping of $(a-3, -b)$ means left 3 and a reflection over x-axis.